

LAB 4 – CONFIGURATION SNMP, SYSLOG and using Wireshark

Install and configure your lab test environment according to the network drawing (Lab 2b) and predefined IP-plan. This lab must be executed on hardware.

Scope definition:

- **Part 1: Configure SNMP**

Enable the SNMP agent feature on both Cisco router and switch and configure the SNMP agent details:

- Use SNMPv2
- Community string read only = SNMP-RO
- Community string read write = SNMP-RW
- SNMP server = 10.199.64.67 (Zabbix server)
- Configure the devices to send SNMP traps and informs:
 - EngineID
 - Enable the device to send SNMP notifications (2 type of notifications are sufficient)
 - Enable syslog trap messages

Reference document that can help to complete the configuration:

[SNMP Configuration Guide - Configuring SNMP Support \[Cisco ASR 1000 Series Aggregation Services Routers\] - Cisco](#)

- **Part 2: Add your devices to Zabbix SNMP-server**

- Connect to the Zabbix SNMP-server and add your device
 - IP address = 10.199.64.67 (http only)
 - User/Pass = 'your classgroup'

Test your configuration:

- Verify SNMP monitoring is working

- **Part 3: Secure SNMP**

- Apply an Access Control Lists (ACLs) to community strings and ensure that the requests community strings are not identical to notifications community strings

Test your configuration:

- Verify your ACL is blocking the traffic to the Zabbix server.

Reference document that can help to complete the configuration:

[Secure Your Simple Network Management Protocol - Cisco](#)

- **Part 3: Enable SYSLOG trap messages**
 - enable syslog messages to the server on 10.199.64.134
 - enable Syslog traps using the snmp-server enable traps syslog command

Reference document that can help to complete the configuration:

[SNMP Configuration Guide - Configuring SNMP Support \[Cisco ASR 1000 Series Aggregation Services Routers\] - Cisco](#)

- **Part 4: Using SNMPv3**
 - Try to add your devices in Zabbix using SNMPv3

Reference document that can help to complete the configuration:

[SNMP Configuration Guide - Configuring SNMP Support \[Cisco ASR 1000 Series Aggregation Services Routers\] - Cisco](#)

- **Part 5: Use wireshark**
 - Try to capture DHCP traffic from a client in VLAN 40 (Data-User segment)

Reference document that can help to complete the configuration:

[Configuring SPAN and RSPAN](#)

[Configuring port mirroring](#)